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Review of Probability Basics

EEE 554 — Lecture #1

Terminology and Notation

- “**Universal outcome set**” S : The set of all possible outcomes of a specified random trial (“experiment”).
- “**Outcome**”: An element of the set S .
- $x \in A$: “ x is an element of the set A .”
- $A \subset B$: “ A is a subset of B .”
- $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$: “The union of A and B .”
- $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$: “The intersection of A and B .”
- $A - B = \{x \mid x \in A \text{ but } x \notin B\}$: “The set difference A minus B .”
- $\emptyset$: “The empty set.”
- $\bar{A} = S - A$: “The complement of A .”

“**Events**”: A collection $\mathcal{F}$ of subsets of S that satisfies the sigma field (σ -field) conditions: 1. $S \in \mathcal{F}$ 2. If $A \in \mathcal{F}$ then $\bar{A} \in \mathcal{F}$ 3. If $A_1, A_2, \dots \in \mathcal{F}$, then $A_1 \cup A_2 \cup \dots \in \mathcal{F}$

Definition: Events $A_1, A_2, \dots$ are **exhaustive** if $A_1 \cup A_2 \cup \dots = S$. They are **mutually exclusive** if $A_i \cap A_j = \emptyset$ for $i \neq j$.

Probability Measure

A probability measure assigns a real number to each event; i.e., $P : \mathcal{F} \rightarrow \mathbb{R}$. It must satisfy: 1. $P(A) \geq 0$ for all events A 2. $P(S) = 1$ 3. If $A_1, A_2, \dots$ are mutually exclusive events,

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

Generation of σ -Fields

Beginning with a collection $\mathcal{B}$ of subsets of S , with $S \in \mathcal{B}$, form every set possible by applying σ -field properties (2) and (3) to the original collection $\mathcal{B}$. The new collection $\mathcal{F}_{\mathcal{B}}$ will be a σ -field and is said to be “generated by $\mathcal{B}$.” If S is a topological space (i.e., with open sets), the **Borel σ -field** is the σ -field generated by the open sets of S .

Conditional Probability

The conditional probability of event A given event B is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Bayes' Rule

From the definition, $P(A \cap B) = P(A|B)P(B)$. But also $P(A \cap B) = P(B|A)P(A)$. So,

$$P(B|A)P(A) = P(A|B)P(B)$$

or

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Statistical Independence

Events A and B are independent if:

$$P(A|B) = P(A)$$

Independent events satisfy $\frac{P(A \cap B)}{P(B)} = P(A)$, so:

$$P(A \cap B) = P(A)P(B)$$

Total Probability Principle

Let $A_1, \dots, A_n$ be mutually exclusive and exhaustive events in S . If B is any event, then $(B \cap A_1), \dots, (B \cap A_n)$ are mutually exclusive events with:

$$B = (B \cap A_1) \cup \dots \cup (B \cap A_n)$$

Thus,

$$P(B) = P(B \cap A_1) + \dots + P(B \cap A_n)$$

or

$$P(B) = P(B|A_1)P(A_1) + \dots + P(B|A_n)P(A_n)$$

Probability of Union

Note that $A = (A - B) \cup (A \cap B)$ with $A - B$ and $A \cap B$ mutually exclusive. So:

$$P(A - B) = P(A) - P(A \cap B)$$

Similarly,

$$P(B - A) = P(B) - P(A \cap B)$$

Finally,

$$A \cup B = (A - B) \cup (B - A) \cup (A \cap B)$$

where the three events on the right-hand side are mutually exclusive. Thus,

$$\begin{aligned} P(A \cup B) &= P(A - B) + P(B - A) + P(A \cap B) \\ &= [P(A) - P(A \cap B)] + [P(B) - P(A \cap B)] + P(A \cap B) \\ &= P(A) + P(B) - P(A \cap B) \end{aligned}$$